

1. Introduction

Althöfer, Haugland, Scherer, Schneider and Van Cleemput [1] introduced alternating plane graphs: plane graphs in which adjacent vertices have different degrees and adjacent faces have different sizes. The condition is severe, and it makes even existence questions hard. Their paper closes ([1], §10, p. 362) with four open problems, of which the first is:

Conjecture 10.1. There are no 2, Y -alternating plane graphs and no X , 2-alternating plane graphs.

Here an X , Y -alternating plane graph has exactly X distinct vertex degrees and exactly Y distinct face sizes. We prove the conjecture (Theorem 4.1); equivalently, every alternating plane graph has at least three distinct vertex degrees and at least three distinct face sizes.

What was known is this. Section 3.2 of [1] shows that a 2, Y -alternating plane graph has degrees 3 and 4, or degrees 3 and 5, and bounds its order below by 25 in the first case and by 56 in the second; the search of its §4 reaches neither. Separately, its §9.1 bounds the number of faces of a *weak* alternating plane graph — one in which vertices of degree 2 are permitted — with degrees 2 and k , by summing $1/r + 1/s$ over the edges separating an r -face from an s -face. That face-incidence count never uses the degree 2. Combining it with the analogous count over vertices excludes both surviving degree pairs at once, and, applied with the roles of vertices and faces exchanged, excludes two face sizes as well.

Why some such combination is needed can be seen immediately. Alternation alone gives at most $\frac{1}{3} + \frac{1}{4}$ from the two endpoint degrees of an edge and at most $\frac{1}{3} + \frac{1}{4}$ from its two incident faces, hence only $\frac{7}{6}$ in total, and $\frac{7}{6} > 1$ is compatible with Euler's formula. The two-valued hypothesis supplies what is missing: it makes the graph bipartite on the side where the two values sit, so on the *other* side the two values are even, hence at least 4 and 6, and that contribution drops to $\frac{1}{4} + \frac{1}{6} = \frac{5}{12}$. The two sides then total exactly 1, and Euler's formula is contradicted. Both halves of the conjecture are this one argument.

The note is organised so that its two achievements stay separate. Section 2 fixes the definitions and the two conventions that the source paper leaves implicit. Section 3 isolates the edge charge and the local parity lemma, with a worked example. Section 4 proves Conjecture 10.1 as stated. Section 5 then proves an extension, not a prerequisite: both nonexistence statements survive when face alternation is required only between *distinct* adjacent faces, so that bridges are admitted. That extension is the longer and harder argument, and its bottleneck is identified there.

2. Preliminaries

We follow [1] throughout. All graphs are finite, simple and plane; those in §3–§5 are also connected. Faces include the unbounded one.

DEFINITION 2.1 ([1], Definition 2.1). An *alternating plane graph* is a connected plane graph in which

- (a) every vertex has degree at least 3;
- (b) every face has size at least 3;
- (c) adjacent vertices have different degrees;
- (d) adjacent faces have different sizes.

DEFINITION 2.2 ([1], Definitions 3.1 and 3.4). A $(3, 4, 5)$ -alternating plane graph is one in which every vertex degree and every face size lies in $\{3, 4, 5\}$. An X, Y -alternating plane graph is one with exactly X distinct vertex degrees and exactly Y distinct face sizes.

CONVENTION 2.3 (face size). The *size* of a face is the length of its boundary walk, that is, the number of edge-side incidences it carries. In particular a bridge, lying on one face from both sides, contributes 2 to that face’s size.

Boundary-walk length is not the number of distinct vertices on the boundary. Two quadrilaterals meeting in exactly one vertex have an exterior boundary walk of length 8, although only 7 distinct vertices lie on it. (This illustrates the convention; it is not an alternating plane graph.) The distinction matters because it is boundary-walk length, not the vertex count, that is forced to be even in a bipartite plane graph.

[1] does not define “size” separately, but its arguments use edge-side incidences: this is the count underlying its equation (3.1) and its face-incidence identity (9.2), and it is what makes the assertion in the proof of its Lemma 9.2 that a bipartite graph has $f_j = 0$ for every odd j correct. Page 339 also treats a bridge as producing a face adjacent to itself, so the paper’s face language is not confined to simple cycles.

CONVENTION 2.4 (bridges). In Definition 2.1, two face occurrences on opposite sides of an edge must have different sizes. A bridge has one face on both sides, so bridges are excluded; [1] says so on p. 339, where an alternating plane graph is noted to be “always at least 2-edge-connected, since plane graph with edge connectivity 1 contains a face that is adjacent to itself”.

Call a plane graph *bridge-relaxed alternating* if it satisfies Definition 2.1 with (d) imposed only on pairs of *distinct* adjacent faces. Bridges are then permitted. Face adjacency is always meant through opposite sides of an edge, never through a shared vertex.

Note that [1]’s “weak” alternating plane graphs relax a different condition, namely (a), by admitting degree 2. The two relaxations are independent, and we use “weak” only in the source’s sense.

3. Edge charge and local parity

We isolate the face-incidence count used in [1], §9.1, and combine it with the analogous count over vertices.

LEMMA 3.1. *Let G be a finite connected simple plane graph with at least one edge, with v vertices, e edges and f faces. For an edge a with ends u, w and face occurrences $F^-(a), F^+(a)$ on its two sides, put*

$$c(a) = \frac{1}{\deg u} + \frac{1}{\deg w} + \frac{1}{|F^-(a)|} + \frac{1}{|F^+(a)|}.$$

Then, by Euler's formula,

$$\sum_a c(a) = v + f = e + 2, \quad \text{equivalently} \quad \sum_a (c(a) - 1) = 2. \quad (1)$$

PROOF. A vertex of degree d is an end of exactly d edges and contributes $1/d$ each time, hence 1 in total; summing gives v . By Convention 2.3 a face of size s occurs exactly s times as a side of an edge and contributes $1/s$ each time, hence 1 in total; summing gives f . $\square$

Neither count requires facial walks to be simple: repeated vertices, cut vertices and bridges are all permitted, a bridge simply having $F^-(a) = F^+(a)$ and contributing $2/|F|$ from its face side. This is what makes Lemma 3.1 usable in §5, where bridges are the object of study. We write $x(a) = c(a) - 1$ for the *excess* of an edge, so that (1) reads $\sum_a x(a) = 2$: some edge must have positive excess.

EXAMPLE 3.2. Start with a triangle and replace each of its three sides by two internally disjoint paths of length two, drawn alongside that side. The result is a simple plane graph with 9 vertices, 12 edges and 5 faces: three quadrilaterals, one for each original side, together with an inner and an outer hexagon. The three original corners have degree 4 and the six new vertices degree 2, so every edge joins a degree-2 vertex to a degree-4 vertex; and every edge lies on one quadrilateral and one hexagon. Hence every edge has

$$c(a) = \frac{1}{2} + \frac{1}{4} + \frac{1}{4} + \frac{1}{6} = \frac{7}{6}, \quad \sum_a x(a) = 12 \cdot \frac{1}{6} = 2,$$

as Lemma 3.1 requires. This graph satisfies both alternation conditions and fails to be an alternating plane graph only through its minimum degree. It also attains $f = \frac{5}{12}e$ with $f = 5$ and $e = 12$, so the face bound $f \leq \frac{5}{12}e$ of ([1], (9.5)) is sharp and cannot be improved to any $\frac{5}{12} - \varepsilon$.

REMARK 3.3. *Run on the weak alternating plane graphs with degrees 2 and k , $k \geq 3$, that [1] treats, the same estimate gives $c(a) \leq \frac{1}{2} + \frac{1}{k} + \frac{5}{12} = \frac{11}{12} + \frac{1}{k}$, which is at most 1 exactly when $k \geq 12$; [1] excludes $k = 11$ by a separate argument (its Lemma 9.4) and exhibits such graphs for $k \leq 10$. This shows where the hypothesis $\deg \geq 3$ enters below. Agreement of the threshold is not an independent verification of the face bound; Example 3.2, which attains the bound, is the check that would detect a strengthening of it.*

The second local ingredient is a parity statement. Stating it over the corners at a vertex, rather than over “the incident faces”, is what makes it correct at a cut vertex, where one face occupies several corners: those corners are non-consecutive, so the alternation below is undisturbed.

LEMMA 3.4 (parity). *Let G be a plane graph whose faces admit a colouring by two colours that is proper on every pair of distinct adjacent faces. Then for every vertex u ,*

$$\deg u - |\{\text{bridges at } u\}| \text{ is even.}$$

PROOF. List the edges at u in cyclic order as $a_1, \dots, a_d$, $d = \deg u$, and let F_i be the face occurrence in the corner between a_i and a_{i+1} , indices modulo d , so that F_{i-1} and F_i are the two sides of a_i . These two occurrences belong to the same face precisely when a_i is a bridge. Hence the corner colour changes across each non-bridge edge at u and is unchanged across each bridge at u . The cyclic list closes up, so the number of changes is even, and that number is $\deg u$ minus the number of bridges at u . $\square$

In particular, if no bridge is present at u then $\deg u$ is even. What the lemma controls is degree minus bridge incidence, not degree itself: in a triangle with a pendant edge the four degrees are 3, 2, 2, 1, the numbers of incident bridges are 1, 0, 0, 1, and the differences 2, 2, 2, 0 are even, although no vertex has even degree. (That graph illustrates the lemma, not Definition 2.1.) In an $X, 2$ -alternating plane graph, and in a bridge-relaxed one, the face size supplies the required colouring.

4. Conjecture 10.1

THEOREM 4.1. *There is no 2, Y -alternating plane graph and no $X, 2$ -alternating plane graph.*

The two cases are the same inequality with the roles of vertices and faces exchanged, but they are not formally dual and we prove both. Passing to the dual graph would need 3-edge-connectivity ([1], Lemma 2.2): a 2-edge cut produces parallel edges in the dual, which therefore need not lie in the simple-graph class we work in. We exchange the two local counting arguments instead.

PROOF. *Case 1: exactly two vertex degrees $3 \leq d_1 < d_2$.* Adjacent vertices differ in degree and only two degrees occur, so every edge joins a d_1 -vertex to a d_2 -vertex. In particular G is bipartite, and every edge contributes

$$\frac{1}{d_1} + \frac{1}{d_2} \leq \frac{1}{3} + \frac{1}{4} = \frac{7}{12}$$

to the vertex side, using $d_1 \geq 3$ from Definition 2.1(a).

Since G is bipartite, every closed walk has even length, so by Convention 2.3 every face has even size, hence size at least 4. By Convention 2.4 the two sides of an edge are distinct faces, and by Definition 2.1(d) their sizes differ; being distinct even numbers

at least 4, one is at least 4 and the other at least 6, so the face side contributes at most $\frac{1}{4} + \frac{1}{6} = \frac{5}{12}$.

Hence $c(a) \leq \frac{7}{12} + \frac{5}{12} = 1$ for every edge, so $\sum_a x(a) \leq 0$, contradicting (1).

Case 2: exactly two face sizes $3 \leq s_1 < s_2$. By Convention 2.4 the two sides of every edge are distinct faces, and by Definition 2.1(d) they have different sizes; with only two sizes available, every edge separates an s_1 -face from an s_2 -face and contributes at most $\frac{1}{3} + \frac{1}{4} = \frac{7}{12}$ to the face side.

The face sizes two-colour the faces, properly on adjacent pairs, and there are no bridges; so by Lemma 3.4 every degree is even. Degrees are at least 3 and even, hence at least 4; adjacent vertices have distinct even degrees, so the vertex side contributes at most $\frac{1}{4} + \frac{1}{6} = \frac{5}{12}$. Again $c(a) \leq 1$ for every edge and (1) is contradicted. $\square$

Case 1 improves ([1], (9.9)) by replacing the degree-2 contribution $\frac{1}{2}$ of its (9.8) by $1/d_1 \leq \frac{1}{3}$; the face bound (9.5) it is combined with is already there, and its derivation never used the degree 2.

5. The bridge-relaxed extension

Definition 2.1 excludes bridges, so Theorem 4.1 is proved. We now show that both conclusions survive the weaker requirement of Convention 2.4, in which face alternation is imposed only between distinct adjacent faces and bridges are therefore admitted.

The relaxation introduces exactly one new difficulty: a bridge need not force a change of face size, so a vertex on a bridge need not have even degree, and the argument of Case 2 above breaks down. The following lemma restores enough control. It says that a bridge is expensive — its face is large — and the proof of Theorem 5.3 then pays for the few edges that can have positive excess out of that expense. The essential step there, and the only place where real work is done, is to bound the number of positive edges assigned to each bridge.

LEMMA 5.1. *In a finite connected simple plane graph of minimum degree at least 3, every face incident with a bridge has boundary-walk length at least 8.*

PROOF. Let $a = uw$ be a bridge and let A and C be the components of $G - a$ containing u and w . Then $\deg_A u = \deg u - 1 \geq 2$ and likewise $\deg_C w \geq 2$, so each of A and C has minimum degree at least 1 and at least three vertices. Let W_A be the facial boundary walk of A through the corner exposed by deleting a . Its length is not 1, which would need a loop. Nor is it 2: that is either a digon of parallel edges, excluded by simplicity, or a single edge traversed in both directions, and for those two darts to constitute the whole facial boundary both ends of that edge would have degree 1, forcing $A = K_2$ against $\deg_A u \geq 2$. So $|W_A| \geq 3$, and likewise $|W_C| \geq 3$. The face F of G incident with both sides of a is traversed as W_A , then a , then W_C , then a in the opposite direction, so by Convention 2.3

$$|F| = |W_A| + |W_C| + 2 \geq 3 + 3 + 2 = 8. \quad \square$$

PROPOSITION 5.2. *There is no bridge-relaxed alternating plane graph with exactly two vertex degrees.*

PROOF. Let G be one, with degrees $3 \leq d_1 < d_2$. As in Case 1 of Theorem 4.1, every edge joins a d_1 -vertex to a d_2 -vertex, so G is bipartite; hence every closed walk has even length — a facial walk traversing a bridge twice included — and every face has even size at least 4. A bridge lies on a face of size at least 8 by Lemma 5.1, so it has

$$c(a) \leq \frac{1}{3} + \frac{1}{4} + \frac{2}{8} = \frac{5}{6},$$

while a non-bridge edge, whose two sides are distinct faces of distinct even sizes at least 4 and 6, has $c(a) \leq \frac{1}{3} + \frac{1}{4} + \frac{1}{4} + \frac{1}{6} = 1$. So $x(a) \leq 0$ for every edge, contradicting (1). $\square$

THEOREM 5.3. *There is no bridge-relaxed alternating plane graph with exactly two face sizes.*

PROOF. Let G be one, with face sizes $s_1 < s_2$. If G has no bridge then it satisfies Definition 2.1 outright and is excluded by Theorem 4.1. So assume G has $B \geq 1$ bridges.

Step 1: only an edge incident with a vertex of degree 3 can have positive excess.

By Lemma 5.1 every face carrying a bridge has size at least 8; since that face is an s_1 - or an s_2 -face and $s_1 < s_2$, this gives $s_2 \geq 8$ as well. Every non-bridge edge separates an s_1 -face from an s_2 -face, so its face side is at most $\frac{1}{s_1} + \frac{1}{s_2} \leq \frac{1}{3} + \frac{1}{8}$. Degrees at the two ends of an edge are distinct by Definition 2.1(c) and at least 3. Three cases:

- a a bridge, its face of size at least 8: $c(a) \leq \frac{1}{3} + \frac{1}{4} + \frac{2}{8} = \frac{5}{6}$, so $x(a) \leq -\frac{1}{6}$;
- a not a bridge, both ends of degree at least 4: $c(a) \leq \frac{1}{4} + \frac{1}{5} + \frac{1}{3} + \frac{1}{8} = \frac{109}{120}$, so $x(a) \leq -\frac{11}{120}$;
- a not a bridge, one end of degree 3: $c(a) \leq \frac{1}{3} + \frac{1}{4} + \frac{1}{3} + \frac{1}{8} = \frac{25}{24}$, so $x(a) \leq \frac{1}{24}$.

Only the third class can be positive. Call its members *positive-eligible* and let P be their number.

Step 2: each bridge receives at most two positive-eligible edges. This is the bottleneck of the proof. Let a be positive-eligible and let u be its end of degree 3, which is unique since adjacent vertices have different degrees. By Lemma 3.4 — applicable because the two face sizes colour the faces properly on distinct adjacent pairs — the number of bridges at u has the same parity as $\deg u = 3$, so it is 1 or 3; it is not 3, since a itself is a non-bridge edge at u . So u lies on exactly one bridge, written $b(u)$, and we assign a to $b(u)$. The assignment is therefore defined on every positive-eligible edge and single-valued. The non-bridge incidence is essential: a degree-3 vertex with three incident bridges would receive no such assignment.

Now fix a bridge b . Every edge assigned to b has its degree-3 end at a vertex whose unique bridge is b , so that vertex is an end of b . The two ends of b have different degrees by Definition 2.1(c), so at most one of them has degree 3; call it u if it exists.

Of the three edges at u one is b itself, leaving exactly two that can be assigned to b . Hence $P \leq 2B$.

Step 3: summing the excesses. Partition the edges into the P positive-eligible ones, the B bridges, and the rest, whose excesses are at most $\frac{1}{24}$, $-\frac{1}{6}$ and 0 respectively by Step 1. Then by (1),

$$2 = \sum_a x(a) \leq \frac{P}{24} - \frac{B}{6} \leq \frac{2B}{24} - \frac{B}{6} = -\frac{B}{12} < 0, \quad (2)$$

a contradiction. $\square$

REMARK 5.4. *The essential point is the bound of Step 2 relating positive-eligible edges to bridges. Even the weaker estimate $P \leq 4B$, obtained by ignoring the degree condition on the two ends of a bridge, would give total excess at most 0 and hence still contradict (1); the bound $P \leq 2B$ supplies the margin $-B/12$ in (2).*

REMARK 5.5. *Theorem 5.3 and Proposition 5.2 are statements about two-valued graphs, not about bridge-relaxed alternating plane graphs in general, which need not be bridgeless. Take three copies H_1, H_2, H_3 of the 17-vertex (3, 4, 5)-alternating plane graph Schneider-17 of ([1], Figure 2), each redrawn so that a face incident with a chosen degree-5 vertex z_i is the exterior face, and join z_2 to z_1 and to z_3 by two bridges drawn in the exterior. Then $\deg z_2 = 7$ and $\deg z_1 = \deg z_3 = 6$, while their neighbours keep degree 3 or 4, so (c) still holds. The merged exterior face has size $|F_1| + |F_2| + |F_3| + 4 \geq 13$, which differs from every interior face size, so (d) holds between distinct adjacent faces. The result is a bridge-relaxed alternating plane graph with bridges. It has neither exactly two degrees nor exactly two face sizes, and so is no counterexample above.*

Disclosure of AI use

Large language models were used in this work and the extent is material. They assisted with exploration, with adversarial review, and with drafting the manuscript; the idea of removing the bridge hypothesis, carried out in §5, was proposed by an AI reviewer. The author re-derived every argument independently, has verified the mathematical content, and is answerable for it and for this disclosure. No AI system is an author.

Computational support

The proofs above are self-contained and depend on no computation. A public repository accompanying this work contains exact-arithmetic checks of the inequalities used here and machine checks of explicit alternating plane graphs against Definition 2.1, including the boundary-walk convention of Convention 2.3. Those checks are advisory and cover examples, not the universal statements proved here; no claim of formal verification is made, and a referee need not run them.

References

- [1] I. Althöfer, J. K. Haugland, K. Scherer, F. Schneider and N. Van Cleemput, *Alternating plane graphs*, Ars Math. Contemp. **8** (2015), 337–363. doi:10.26493/1855-3974.584.09a.

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